

TCS NQT PYQ QUESTIONS

Company: TCS NQT

Round: Online

Year: 2025

Difficulty: Medium-Hard

Topic: Probability

Problem Description:

A box contains 5 red, 4 green, and 3 blue balls (12 balls in total). Three balls are drawn at random, without replacement.

What is the probability that at least two of the three balls drawn are of the same color?

Options:

- A) 5/11
- B) 7/11
- C) 8/11
- D) 9/11

Correct Answer:

C) 8/11

Approach to Solve This Problem:

STEP 1:

It is easier to work with the complementary event: all three balls drawn have different colors (one red, one green, one blue).

Total ways to choose any 3 balls from 12 = $C(12,3) = 220$.

STEP 2:

Ways to pick one ball of each color = $5 \times 4 \times 3 = 60$.

So $P(\text{all three different colors}) = 60/220 = 3/11$.

STEP 3:

$P(\text{at least two the same color}) = 1 - P(\text{all different}) = 1 - 3/11 = 8/11$.

CORRECT ANS = 8/11